



Norwegian Startup 'N2 Applied'

Promoting Sustainable Agriculture Through Technology

Case Study

This case was written by **Nilosha Sharma** and reviewed by **K. Bhagyalakshmi**, Amity Research Centers Headquarter, Bangalore. It is intended to be used as the basis for class discussion rather than to illustrate either effective or ineffective handling of a management situation. The case was compiled from published sources.

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Author: Nilosha Sharma

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Case Study

*"We are proud that our technology is part of this collaboration. Industry, farmers and consumers recognise the need to make food production as sustainable as possible. Our plasma technology provides significant reductions in nitrogen pollution and greenhouse gas emissions, which are two major challenges facing agriculture. At the same time, the resources on the farm are utilised in a much better way."*¹

– Carl Hansson, CEO, N2 Applied

One of the most important concerns across the globe was sustainable food production. It was the need of the hour to build efficient and sustainable food systems. But it required new agriculture and aquaculture technologies. One of the Nordic countries, Norway realised that Earth's resources were limited and thus strived to work towards creating responsible and sustainable value chains.² Meanwhile, an international agricultural technology business named as N2 Applied based in Oslo, Norway aimed to advance its priorities in perfecting global food production by reducing emissions and increasing yields. N2 Applied created a technology that facilitated local production of fertilisers using livestock or slurry, air and electricity.³

Founded in 2010 by Rene Ingels and Grete Sønsteby⁴, N2 Applied was 'instrumental in achieving climate targets on an industrial scale'.⁵ N2 Applied made advances in achieving sustainable development goals which included zero hunger, good health and well-being, Industry, innovation and infrastructure, responsible consumption and production and climate action. For this, N2 Applied made significant investments in resources in field trials from

¹ "N2 Applied's technology will be used by food industry giant", <https://n2applied.com/2023/08/31/n2-applieds-technology-will-be-used-by-food-industry-giant/>, August 31st 2023

² "Sustainable agriculture and food production with Norwegian technology", <https://businessnorway.com/articles/sustainable-agriculture-and-food-production-with-norwegian-technology>, March 9th 2023

³ "N2 Applied and GEA enter strategic partnership in sustainable dairy farming", <https://n2applied.com/2022/11/16/n2-applied-and-gea-enter-strategic-partnership-in-sustainable-dairy-farming/>, November 15th 2022

⁴ "FOUNDERS", <https://n2applied.com/founders/>

⁵ "N2 Applied reduces emissions from food production and turns organic waste into an effective fertiliser", <https://businessnorway.com/solutions/n2-applied-reduces-emissions-from-food-production-and-turns-organic-waste-into-an-effective-fertiliser>, November 28th 2022

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Northern Norway to Africa. With N2 technology in place, farmers were able to produce more for the fast growing population in a sustainable way.

Sustainable Agriculture and Food Production in Norway: An Overview

Producing sustainable food was the key concern for countries across the globe. According to a UN report in 2017, “One in nine people in the world went to bed hungry, while one in three suffered from malnutrition.”⁶ According to Food and Agriculture Organisation (FAO), of the United Nations, “Each year, the world loses or squanders a third of the food it produces. This means that somewhere between planting seeds in fields and providing nourishment to the world’s 7 billion people, approximately 1.3 billion tonnes of food with a value of more than US\$1 trillion is lost or wasted. Food and Agriculture Organization (FAO) of the United Nations is urging that countries do their best to keep trade flowing and food supply chains alive and increase agricultural output during the international health crisis.”⁷

In the meantime, the resources on earth were inadequate and were not enough to fulfill the food needs all individuals. To fulfill these needs, efficient agriculture and aquaculture technology was needed so as to build responsible and sustainable value chains. Seeing the growing need for sustainable agriculture and food production in the country, Norway had taken several initiatives.

Also, Norwegian producers had to play a crucial task in smart and efficient food production technologies for achieving the targets of the Paris Agreement. Norway developed modern and technologically advanced food production systems in the country. Since the 1960s, the land-based food production almost doubled in Norway. Due to the use of modern technology, the labour input had reduced to almost 35%. For example, the dairy farmers of Norway were far more advanced in using robots for milking. As Norway was using advanced technology, it helped the farmers to improvise animal welfare and sustainable food production which ultimately saved time and money. This resulted in more number of young people joining agriculture.

Furthermore, to achieve sustainable food production there was remarkable public engagement. Improvised food transparency and traceability were the need of the hour, and one of the means for achieving this was blockchain technology. Many companies in Norway worked towards creating sustainable food production food waste reduction. One such company was KeepIt Technologies which came up with Keep-it indicator which constantly monitored the temperature of the product and indicated the actual shelf life of a product.⁸ The idea of Keep-it was developed by four professors at the University of Life Sciences in Norway.⁹ (Exhibit I).

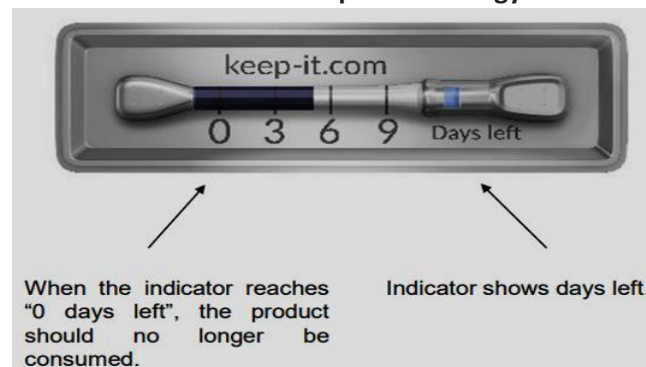
⁶ “Sustainable agriculture and food production with Norwegian technology”, op.cit.

⁷ “TotalCtrl: Food Waste Management Technology Startup: Making the Food Value Chain Sustainable”, <https://vijaykrishnacv.com/2020/05/04/totalctrl-food-waste-management-technology-startup-making-the-food-value-chain-sustainable/>

⁸ “Sustainable agriculture and food production with Norwegian technology”, op.cit.

⁹ Faye Therese and Pettersen Marit Kvalvåg, “Keep-it® time- and temperature indicator”, https://www.actinpak.eu/wp-content/uploads/2018/11/Vienna_presentation_marit_kvalvag_pettersen.pdf, 2018

Exhibit I Features of Keep-it technology



Source: Faye Therese and Pettersen Marit Kvalvåg, "Keep-it® time- and temperature indicator", https://www.actinpak.eu/wp-content/uploads/2018/11/Vienna_presentation_marit_kvalvag_pettersen.pdf, 2018

The Keep-it indicator was custom made according to the requirement of each product and then was attached to the food package at the production line in the manufacturing unit.¹⁰ (Exhibit II).

Exhibit II Custom Made Keep-it Indicator

Indicator categories: - Chilled products - Freeze products - Freeze - thaw products	Tailormade graphical design: - Countdown of days - Fresh - Expired - Use - Do not use - OK - Not OK - Best before (indicated with simple line) - Frozen - Ready - Expired
Indicator specs: - Shelf life : From 2 to 30 days - Packages: MAP, skinpack, flowpack & vacuum pack - Stored & transported chilled - Shelf life storage : 3 months	

Source: Faye Therese and Pettersen Marit Kvalvåg, "Keep-it® time- and temperature indicator", https://www.actinpak.eu/wp-content/uploads/2018/11/Vienna_presentation_marit_kvalvag_pettersen.pdf, 2018

Another startup, TotalCtrl also took upon the challenge to solve the food waste problem. TotalCtrl was founded by Charlotte Aschim (Aschim), as food waste management company. The company offered software/technology solutions to reduce food waste. Aschim stated, "We are a Software company. We have solutions available for Grocery Stores, Food Banks, Restaurants & Municipalities. We are helping our customers to save time. Instead of manually checking the inventory / stock in Grocery Store which are about to expire, we help in automating it. It also helps to give the indicators to move such products to market place / reduce the price & try to sell it instead of getting wasted. For Restaurants, they have multiple storages. It is an extremely traditional industry. We help them to save around 85% of food waste & save lot of time & money. So we help in making a huge impact in our customers' profits as well as the planet & people in general."¹¹

¹⁰ "Keep-it® time- and temperature indicator", op.cit.

¹¹ "TotalCtrl: Food Waste Management Technology Startup: Making the Food Value Chain Sustainable", op.cit.

Besides, to create safe and healthy Norwegian food, the government of Norway came up with a broad political framework 'Food Nation Norway' for business development and value creation. As part of Food Nation Norway initiative organisations and individuals had to 'commit themselves to an action plan' which focused on 'good plant health', 'good animal and fish welfare' and 'documentable quality of all segments of the food chain'. The aim of this Food Nation Norway initiative was to attain 'sustainable food production and use of food', 'sustainable agriculture', 'exceptional food and drink experiences', and 'reduction of food waste as per national and international commitments by 2030'.¹²

Furthermore, the Food Nation Norway strategy was developed for positioning Norway as a Food Nation globally and visible marketing element in promoting Norway as a tourist destination especially for its food culture, for its large exports of seafood, and for its food and drinking experiences. The strategy of government included four action areas to achieve the vision of Norway. Firstly, it included production and offering of sustainable, safe and healthy food. Secondly the strategy should create the basis for growth and value creation across the country. Then there was innovation and diversity, and competence, recruitment and reputation building.¹³ (Exhibit III).

Exhibit III
Food Nation Norway Initiative – Four Action Areas

Sr.No.	Initiatives	Description
1	The production and offering of sustainable, safe, and healthy food of high quality	Pursue a comprehensive business and health policy that emphasizes sustainability, good health and safe and healthy food of high quality; Continue monitoring the Norwegian diet; Update and renew the national strategy against antibiotic resistance; Strengthen collaboration with the business community for a healthier diet; etc.
2	Creating the basis for growth and value creation throughout the country	Conduct an active business policy in the area of agriculture and seafood; Develop a strong supply-sector of technologies and services; Continue working towards building strong regions for food and tourism; Stimulate a more active use of food-labelling schemes for added value and attractive regions for food and tourism; etc.
3	Innovation and diversity	Continue to prioritize business-oriented measures for food and tourism; Support the development of comprehensive innovation systems in the area of food that can respond to societal needs; Strengthen the link between public procurement needs and the food and catering services; etc.
4	Competence, recruitment and reputation building	Contribute to an increased knowledge among children and young people about a healthy and sustainable diet, the use of ingredients, practical cooking, food and food culture; Contribute to increased knowledge in the population about sustainable food production in cities and settlements; Work for increased competence and a strengthened recruitment in the food and catering industry; Contribute to building a comprehensive reputation that strengthens, further develops and positions Norway as a Food Nation nationally and internationally.

Source: "Food Nation Norway", <https://www.fao.org/faolex/results/details/es/c/LEX-FAOC202977/>

¹² "Sustainable agriculture and food production with Norwegian technology", op.cit.

¹³ "Food Nation Norway", <https://www.fao.org/faolex/results/details/es/c/LEX-FAOC202977/>

Moreover, the government of Norway launched a new strategy on food security named as 'Gathering power against famine – a policy for increasing self-sufficiency'. The plan focused on pushing 'Norwegian efforts on food security in low and middle income countries'. In addition to this, the plan also 'supported local and regional, small-scale producers of food, local job creation in the value chains, and access to nutritious food'.¹⁴ (Exhibit IV).

Exhibit IV
Four Priority Areas of Norway

Sr.No.	Priority Areas
1	Improving productivity, including reduced production losses, for small-scale producers of food
2	Developing robust local value chains and markets that strengthen the position of small-scale producers of food and small and medium-sized enterprises (SMEs)
3	Increasing and improving access to a healthy and safe variety of foods; and
4	Addressing the root causes of food insecurity and building partner countries' capacity to respond to natural and social barriers to food security

Source: "Norway launches new strategy for global food security",
https://donortracker.org/policy_updates?policy=norway-launches-new-strategy-global-food-security#all-updates, November 29th 2022

In addition to this, Norway also set few objectives to attain higher goals for the country (Annexure I). Along with establishing objectives, the government of Norway joined hands with 'Ghana, Malawi, Mozambique, South Sudan, and Tanzania'. The particular countries were selected 'based on the food security challenges faced by the countries' and the 'existing Norwegian development infrastructure in the states'. Other than this, Norway launched a new agenda on food security which included several key partners like 'civil society organizations that work directly with food producers', 'farmers' and fishermen's organizations that provide services to producers', and 'private companies working with farmers provide inputs or refine products'.¹⁵

Apart from national objectives and targets, Norwegian companies used more precision agriculture methods and digitisation to increase production. For example, Agritech covered a range of agriculture and technological innovations. Other than this, Robotics and AI helped in farming and brought in major advances in precision. In Norway, robots changed the whole scenario of farming as they helped to reduce herbicide use and even solved manual labour shortage. For example, Kilter's AX-1, was an 'autonomous weeding robot that sprays herbicide droplets directly onto weeds, without hitting crops or soil'. The AX-1 robot made use of deep learning neural networks, so as to work in the field with patented nozzle technology thereby reducing the use of herbicide by 95%. The robot was lightweight in design so as to reduce the unwanted soil compaction.

Anders Brevik, CEO of Kilter, stated, "Kilter comes from the American saying 'out of kilter', which means out of balance. We chose the name because we want to bring balance back to Mother Nature. We want to be a part of the shift towards a better use of natural resources and hopefully contribute by giving back to nature." In addition to this, another Norwegian Agritech innovation was Thorvald from Saga Robotics which was a 'platform for autonomous multifunctional robots' (Annexure II). The Thorvald robot was used for 'picking fruit and

¹⁴ "Norway launches new strategy for global food security",
https://donortracker.org/policy_updates?policy=norway-launches-new-strategy-global-food-security#,
November 29th 2022

¹⁵ *ibid.*

vegetables to cutting grass for forage, performing UV treatments and collecting and analysing data for crop predictions’.

In addition to this, Thorvald could be easily assembled in different geography and in different types of crops in field. The robot was lightweight and did not compress the soil. Pal Johan From (Johan From), Chief Product Officer of Saga Robotics, viewed that, “We are only beginning to see the tremendous impact Agritech will have in the coming years’. Johan From stated, “I passionately believe robotics can make an enormous positive difference to farmers. Technology is allowing an ever-growing number of farm tasks to be easily transferred to robots, freeing up time and resources to focus on work that requires human involvement. The results are encouraging, with customers saving significant amounts of labour and improving yields.”

Apart from smart farms and robots, Norway worked upon improving the food supply chain through data analysis. For example, digital solutions company Mimiro promoted profitable food production through various digital solutions for farmers. Mimiro also had useful apps which helped ‘farmers to optimise production’, ‘increase income across the value chain’, and ‘reduced the greenhouse gas emissions’. Along with this, Mimiro was also working towards creating a data platform and machine learning services so that it could be used across by multiple stakeholders **(Annexure III)**.

Another example was of Farmforce, which provided farmers with digital solutions so as to improvise their lives, protect the environment and ensure sustainable sourcing. Further the platform helped digital management of small holder farming schemes, and also looked after everything from production to the point of distribution. Arnaud Dupuis, Marketing and Communications Manager, Farmforce, stated, “Our motto is data with a purpose. Data can’t solve anything if it just remains data. We are working to give life to the data and to transform it into actionable insights for our customers so they can take that insight and make good and sustainable decisions.”

Besides, Agritech also helped to meet the UN’s 17 Sustainable Development Goals (SDGs). For example, GreenCap Solutions worked towards Zero Hunger and Responsible Consumption and Production with the help of environmental control system (ECS) which it had created for industrial greenhouses. The services provided by GreenCap ECS included helping producers to grow vegetables with carbon-neutral footprint, energy-efficient climate control, and also reduced freshwater consumption. Jarle Skjæveland, CSO of GreenCap Solutions, stated, “We’ve seen the United Nations estimates which state that food production needs to increase. We believe our system can play a pivotal role in reaching that goal in a sustainable way.”

The process of GreenCap Solutions¹⁶ was a closed-loop system which ‘saved energy by capturing carbon dioxide from the atmosphere’ and then ‘reused it inside the greenhouse’. Then the captured CO₂ was circulated inside the greenhouse such that the CO₂ concentration was optimal for growth. This ended up increasing the growth rate of plant by atleast 40%. Other than this, GreenCap system used up to 70% of water consumed in production. According to McKinsey, “Those who embrace the change at an early stage may be best positioned to thrive in agriculture.”¹⁷ Likewise, another company named N2 Applied, provided on-farm system which helped farmers make fertiliser using locally sourced manure and renewable energy.¹⁸

¹⁶ Environmental technology provider of large-scale carbon dioxide removal solutions.

¹⁷ “Norwegian agritech is transforming farming through digitalisation”, <https://businessnorway.com/articles/norwegian-agritech-is-transforming-farming-through-digitalisation>, March 9th 2023

¹⁸ “Sustainable agriculture and food production with Norwegian technology”, op.cit.

Can N2 Applied Transform Norway's Agri Sector through Sustainable Solutions?

N2 Applied was founded in 2010¹⁹ by Rune Ingels (Ingels) and Grete Sønsteby (Sønsteby). Ingels worked for Hydro and Yara in production and in technology strategy and R&D. In 2010, Ingels left his position in Yara and started to adopt his own technology ideas. He along with Sønsteby cofounded N2 Applied which worked on green nitrogen technology.²⁰ N2 Applied addressed two problems faced by farmers. The farmers had a task to feed many more people and at the same time reduce greenhouse gas emissions. Carl Hansson (Hansson), CEO of N2 Applied, stated, "Current agricultural practices result in a loss of nutrients, significant emissions and the use of chemical fertilisers based on fossil fuels. This needs to change."

To deal with these circumstances, N2 Applied facilitated farmers to 'turn liquid manure also known as slurry into sustainable, low emission fertiliser with the help of only air and electricity'. Hansson stated, "We are reducing agricultural emissions with the farmers' own resources. Now we have a nitrogen-rich fertiliser that is better for the farmer, better for the planet and better for all of us." The slurry contained rich nutrients and also had fertilisation properties which could be used. With traditional farming practices, these nutrients were lost and ended up in pollution of air, water and ecosystems.

Other than this, organic waste also released greenhouse gases like methane and nitrous oxide which was not only challenging for the environment but also for farmers. Further to maintain yield, there was need for chemical fertiliser but then again it emitted greenhouse gas. Therefore, farmers and Agri-food industry looked for solutions which increased both sustainability and profitability.

Thus, by using N2 unit of N2 Applied, farmers were able to harvest their own fertiliser from livestock slurry, air and electricity. N2 unit added 'nitrogen from the air into the slurry which ultimately increased the nitrogen content'. The general process was while spreading slurry into the fields much of the nitrogen was evaporated into the air as 'ammonia or leaches into the soil as nitrates'. Hansson stated, "Firstly, it improves organic fertiliser by stopping the loss of ammonia and increasing the fertiliser value. This optimises crop production. Secondly, it stops methane emissions and prevents ammonia emissions, which cause global warming and reduce air quality. This can all be done using only manure, air and electricity – turning the nitrogen green in several ways." **(Watch the Video of how the N2 Technology worked – <https://www.youtube.com/watch?v=p8aJH6Vsfa>).**

Then the end product was the nitrogen enriched organic fertiliser (NEO) which was same as normal slurry but comprised of more nitrogen and less emissions. This helped farmers to spread the same fertiliser using 'existing farm equipments thereby improvising their own food production', 'reduce the need for chemical fertiliser and making farming more circular'.²¹ The machines size of N2 Applied was that of shipping container and could be easily fitted into the infrastructure. Along with this, the 'liquid manure was treated continuously prior to storage for application at the right time'. The farmers could set up the machine between the livestock shed and the manure storage.

¹⁹ "N2 Applied", <https://no.linkedin.com/company/n2applied>

²⁰ "FOUNDERS", op.cit.

²¹ "N2 Applied reduces emissions from food production and turns organic waste into an effective fertiliser", op.cit.

Apart from livestock farming, 'N2 plasma technology used different forms of liquid matter as feedstock and had various applications'. Also it was used in sectors like 'arable farming, aquaculture and biogas'. By combining biogas and the N2 technology, one could increase the 'environmental benefits by upgrading the bi-product', 'biogas digestate', 'to power nitrogen fertiliser and stopping the remaining methane production and ammonia emissions'. Other than this, N2 Applied offered actual solutions to meet the sustainability targets and demand of the consumers. N2 Applied could further be used for ammonia scrubbing, making artificial proteins and create fertilisers from waste streams, dust and ashes. The N2 plasma technology was a new method to reduce ammonia and greenhouse gas emissions while increasing the nitrogen content in the organic material.²² **(Exhibit V).**

Exhibit V
The N2 Plasma Technology

Sr.No.	Process	Description
Step 1	Plasma Treatment	Electricity is used to split the nitrogen (N ₂) – and oxygen (O ₂) molecules in air. The atomic nitrogen (N) and oxygen (O) then recombine and form a reactive nitrogen gas.
Step 2	Absorption	The reactive nitrogen gas is absorbed into the liquid phase of an organic material, e.g. livestock slurry or biogas digestate, and is converted to plant available nitrogen

Source: "The N2 Plasma Technology", <https://n2applied.com/the-technology/>

In addition to this, N2 Applied was involved in R&D activities and technology development focus was mainly on four areas which included Plasma, absorption, agronomy and cloud & control systems **(Annexure IV)**. Thereafter in 2021, the European Commission's European Innovation Council (EIC) granted €15 million finance as equity funds to N2 Applied. Hansson stated, "It is becoming more and more important to produce food with fewer emissions, and our solution goes hand in hand with the targets the World has set. We plan to commercialise the technology, and this announcement gives us the power and speed to move ahead towards commercialisation."²³

To further accelerate and scale the adoption of N2 Applied technology, the company made partnership with GEA which provided solutions to farms and the dairy industry. The aim of the partnership was to provide solutions for sustainable livestock farming in the global market and to enable dairy farmers to reach their sustainability targets. Hansson stated, "Joining forces with GEA Farm Technologies puts our technology onto a global stage and can enable us to drive its adoption by livestock farmers much faster. At a time when farming is under pressure from environmental challenges and cost of chemical fertilisers have seen large price increases combined with a scarcity of supply, the partnership will offer a clear and compelling path to more assured and sustainable food production."²⁴

Thereafter in 2023, N2 Applied raised around €10 million in investment funding so as to further improvise its fertiliser technology and expand its commercial use.²⁵ Apart from raising

²² "The N2 Plasma Technology", <https://n2applied.com/the-technology/>

²³ "N2 Applied raises €15 million of funding from the EU", <https://n2applied.com/2021/10/15/n2-applied-raises-e15-million-of-funding-from-the-eu/>, October 15th 2021

²⁴ "N2 Applied and GEA enter strategic partnership in sustainable dairy farming", op.cit.

²⁵ "N2 Applied Raises €10 Million / NOK 105 Million to Expand the Use of Breakthrough Plasma Treatment Technology for Sustainable Fertiliser Production", <https://n2applied.com/2023/01/11/n2-applied-raises-e10-million-nok-105-million-to-expand-the-use-of-breakthrough-plasma-treatment-technology-for-sustainable-fertiliser-production/>, January 11th 2023

funds, the partnership of N2 Applied with GEA proved to be fruitful for the company as GEA partnered with Unilever to improvise sustainability on dairy farms. Initially, GEA 'installed four units of the ProManure E2950 Manure Enricher', to further use N2's plasma technology on 'Dutch farms that supplied milk to Unilever'.²⁶

By using N2 Applied technology, farmers were able to enhance crop yields by 40%, thereby fulfilling the Sustainable Development Goals (SDGs) 2 of Zero hunger which was about ending the hunger of people and achieving food security and improved nutrition. Also, N2 Applied reduced down greenhouse gas emissions, thereby achieving the SDG goal 13 which worked upon combating climate changes and its impacts. Hansson stated, "Our technology will be instrumental in achieving climate targets on an industrial scale. Not many solutions can do that holistically."

Besides, N2 Applied also interrupted the fertiliser value chain turning it from a 'linear to a circular model'. This technology helped to ensure responsible consumption and production of food, SDG goal 12. By producing responsibly, N2 Applied fulfilled the SDG goal 3 which ensured healthy life. Other than this, N2 technology helped farmers to produce their own nutrient rich fertiliser thereby reducing their reliance on costly industrial fertilisers. This led to achieving SDG goal 9 of building resistant infrastructure, promoting comprehensive and sustainable industrialisation and foster innovation. By providing better means production, N2 Applied ensured farmers were able to produce more food for increasing global population in a sustainable way.

Hansson stated, "Farmers are not short-term thinkers. They have a 20 to 30-year perspective and look for long-term gains. Food companies, from local retailers to large corporations, also have incentives to reduce agricultural emissions. They aren't doing this to be nice. They are responding to consumer and shareholder demands." Furthermore, N2 Applied had received from 'the European Innovation Council (EIC) Accelerator of €17.5 million' and also had 'machines in the Netherlands, Norway, Sweden and the UK'. Hansson stated, "In recent years, we have invested large sums of money and resources in field trials from Northern Norway to Africa. The research shows that our N2 Unit significantly cuts emissions and increases crop yields."²⁷

Apart from this, in 2024, a new study was conducted by Norwegian University of Life Sciences (NMBU) which showcased how by using N2 plasma technology, methane formation in stored livestock could be eliminated completely. The study also compared the 'methane formation from cow manure to the methane formation in nitrogen enriched cow manure (NEO)'.²⁸

N2 Applied joined hands with Felleskjøpet Agri²⁹, in an attempt to offer a sustainable fertiliser solution on Norwegian farms. The companies signed a two-year letter of intent to provide

²⁶ "N2 Applied's technology will be used by food industry giant", <https://n2applied.com/2023/08/31/n2-applieds-technology-will-be-used-by-food-industry-giant/>, August 31st 2023

²⁷ "N2 Applied reduces emissions from food production and turns organic waste into an effective fertiliser", op.cit.

²⁸ "New study from Norwegian University of Life Sciences (NMBU) shows complete elimination of methane formation in stored livestock manure using N2's plasma technology", <https://n2applied.com/2024/01/31/new-study-from-norwegian-university-of-life-science-nmbu-shows-complete-elimination-of-methane-formation-in-stored-livestock-manure-using-n2s-plasma-technology/>, January 31st 2024

²⁹ It is one of the main cooperatives in Norwegian agriculture.

farmers with the N2 technology and help them in recycling nutrients and reducing emissions of greenhouse gases and ammonia.³⁰

Apart from this, talking about the risk strategy, Rasmus Sunde, CEO, N2 Applied, stated, “You have a commercialization risk linked to going from having nine pilots installed to the delivery of 10,000 machines by 2030. Then there’s the risk that a serious competitor might arrive, although we don't currently see many of them at the same level of maturity as us. You also have the risk of getting a new technology embraced by the market. I think the main other risk would be that nothing happens on the regulatory side. Denmark is a leader in the race to regulate emissions, as the Danish realize the agriculture sector is a main contributor to greenhouse gases. I think the EU will push for regulation that incentivizes farms to deploy sustainable technologies, such as ours.”³¹

Amidst this scenario, would N2 Applied help in reducing agriculture emissions and provide viable solutions to ensure sustainability and trade profitably?

Annexure I Intermediate Objectives of Norway Government

Sr.No.	Objectives
1	Identify and build relevant partnerships.
2	Support national plans and strategies for food security, sustainable development, and development of agriculture, fisheries and farming.
3	Work to ensure harmony between different thematic initiatives.
4	Facilitate increased cooperation with civil society and the private sector.
5	Conduct follow-up research and impact evaluation to ensure positive, concrete results.
6	The strategy will inform Norwegian efforts through the UN, development banks, and global funds, and will be used as a basis for bilateral commitments on food security in partner countries.

Source: “Norway launches new strategy for global food security”,
https://donortracker.org/policy_updates?policy=norway-launches-new-strategy-global-food-security#all-updates, November 29th 2022

³⁰ “Felleskjøpet Agri and N2 Applied enters agreement”, <https://n2applied.com/2024/03/19/felleskjopet-agri-and-n2-applied-enters-agreement/>, March 19th 2024

³¹ “Rasmus Sunde (N2 Applied): ‘Our solution helps farmers reduce emissions and improve productivity’”, <https://www.leadersleague.com/en/news/rasmus-sunde-n2-applied-our-solution-helps-farmers-reduce-emissions-and-improve-productivity>, December 20th 2022

Annexure II

The Thorvald Robot of Norway



Source: "Norwegian agritech is transforming farming through digitalisation", <https://businessnorway.com/articles/norwegian-agritech-is-transforming-farming-through-digitalisation>, March 9th 2023

Annexure III

Farmable's Farming App



Source: "Norwegian agritech is transforming farming through digitalisation", <https://businessnorway.com/articles/norwegian-agritech-is-transforming-farming-through-digitalisation>, March 9th 2023

Annexure IV

Technology Development and R&D at N2 Applied

Areas	Description
Plasma	Plasma Technology is a core part of the N2 process. Optimising the reactor for reactive nitrogen production is a main R&D focus.
Absorption	The absorption process is where reactive nitrogen is converted to quality fertilisers. Developing efficient absorption technology is a core R&D area.
Agronomy	Aimed at validating the NEO fertiliser and applications.
Cloud and Control Systems	Software for controlling the machines and R&D equipment, and cloud for data collection, visualisation and analytics.

Source: "The N2 Plasma Technology", <https://n2applied.com/the-technology/>